



PROPOSAL PRESENTATION

CSE 4502: Operating System Lab

Team : 09

TEAM MEMBERS

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WHAT WE ALREADY HAVE

An on-development OS

W I T H

- Bootloader & Physical Memory Management
- Container & Virtual Memory Management
- Process Management & Trap Handling
- Multicore & Preemption
- File Systems



PROJECT GOAL

TO IMPLEMENT A FUNCTIONAL UNIX SHELL on mcertikOS

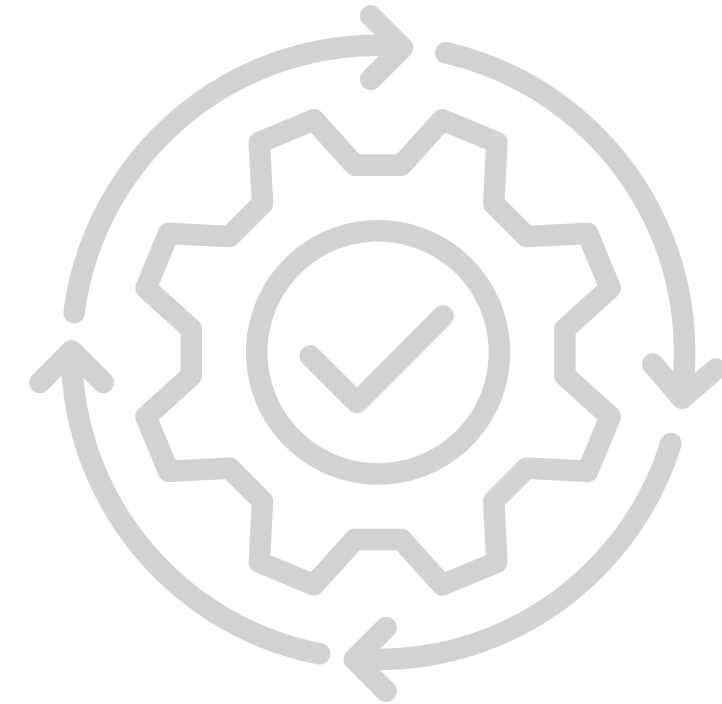
W I T H

- USER PROCESS SEPARATION
- UNIFIED FILE DESCRIPTORS
- REDIRECTION & PIPING
- EXTENSIBILITY & SCALABILITY

IMPLEMENTATION PLAN

FEATURE 01

Process-based Command Execution



- Modify `spawn()` to create separate user-space processes for each command.
- Ensure processes inherit correct file descriptors for I/O redirection.

IMPLEMENTATION PLAN

FEATURE 02

File Descriptor Abstraction

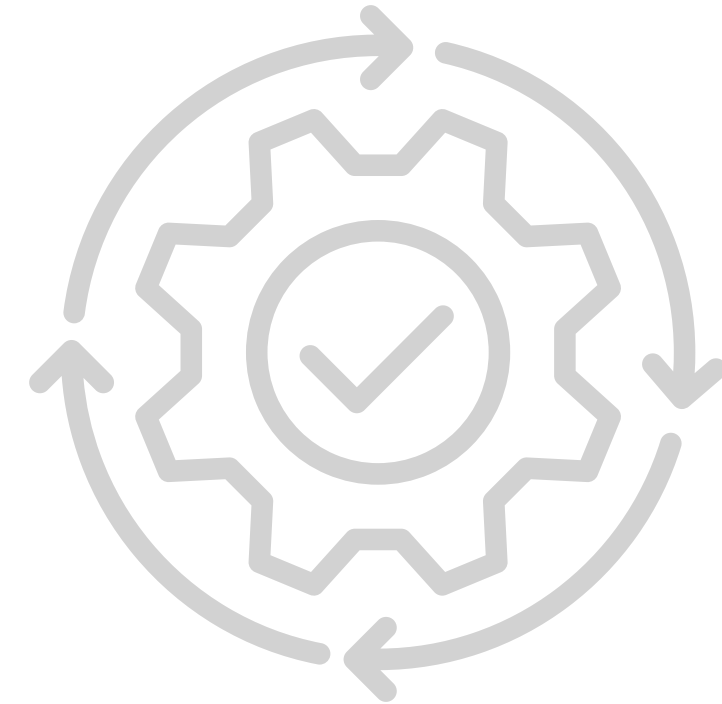


- New system call to create file descriptors for:
 - Console Input (stdin)
 - Console Output (stdout)
 - Pipes (IPC)
 - Files (read/write from disk)
- Store file descriptors as integers in userspace for easy reference.

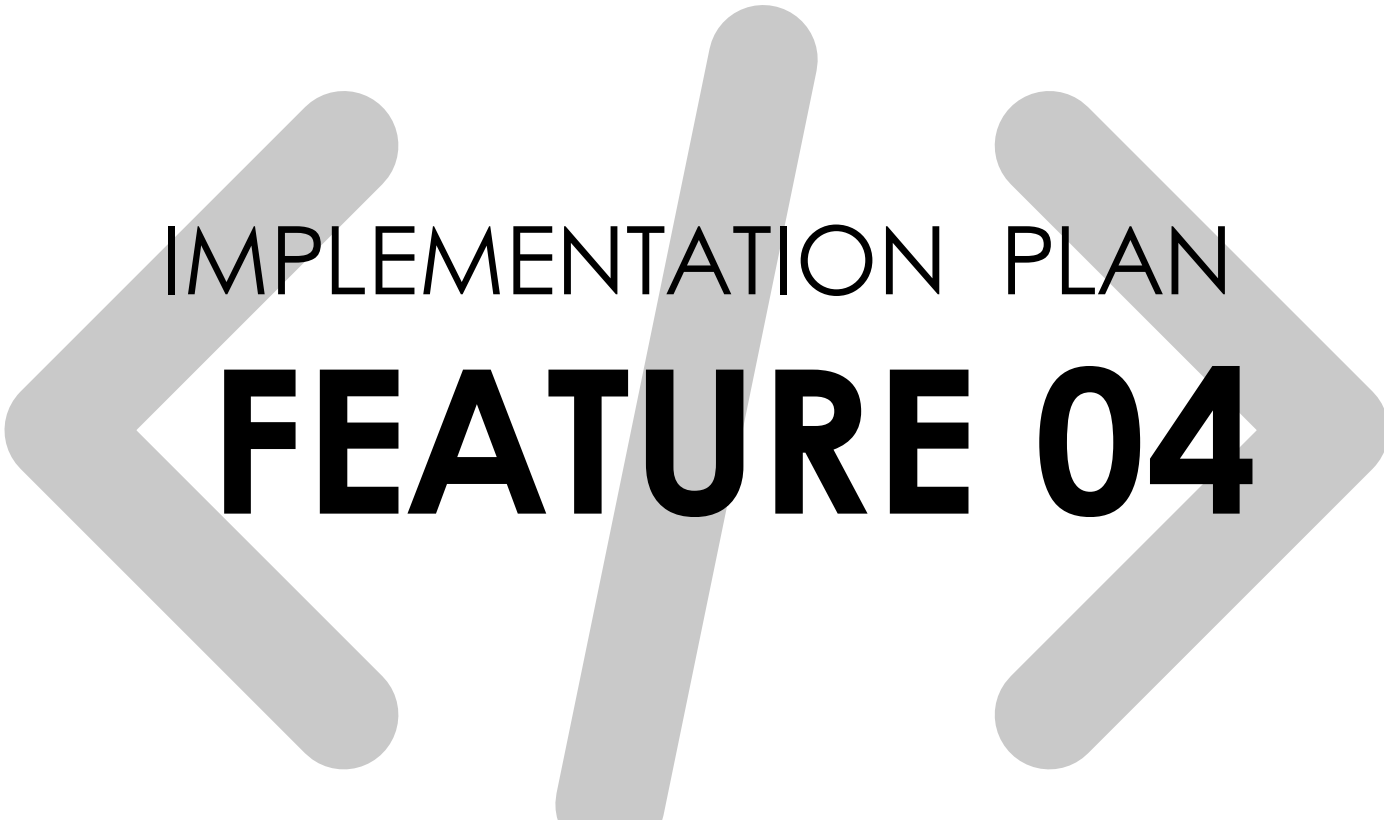
IMPLEMENTATION PLAN

FEATURE 03

System Call
Overhaul



- Modify the kernel syscall interface to handle read and write calls properly.
- All user processes must use file descriptors instead of direct console I/O.



IMPLEMENTATION PLAN

FEATURE 04

Pipes & Redirection



- Modify the shell parser to recognize |, <, and >.
- Use pipes (pipe() syscall) to create inter-process communication channels.
- Redirect file descriptors when spawning new processes (spawn(cmd, in_fd, out_fd)).

IMPLEMENTATION PLAN

FEATURE 05

Execute Processes from Disk



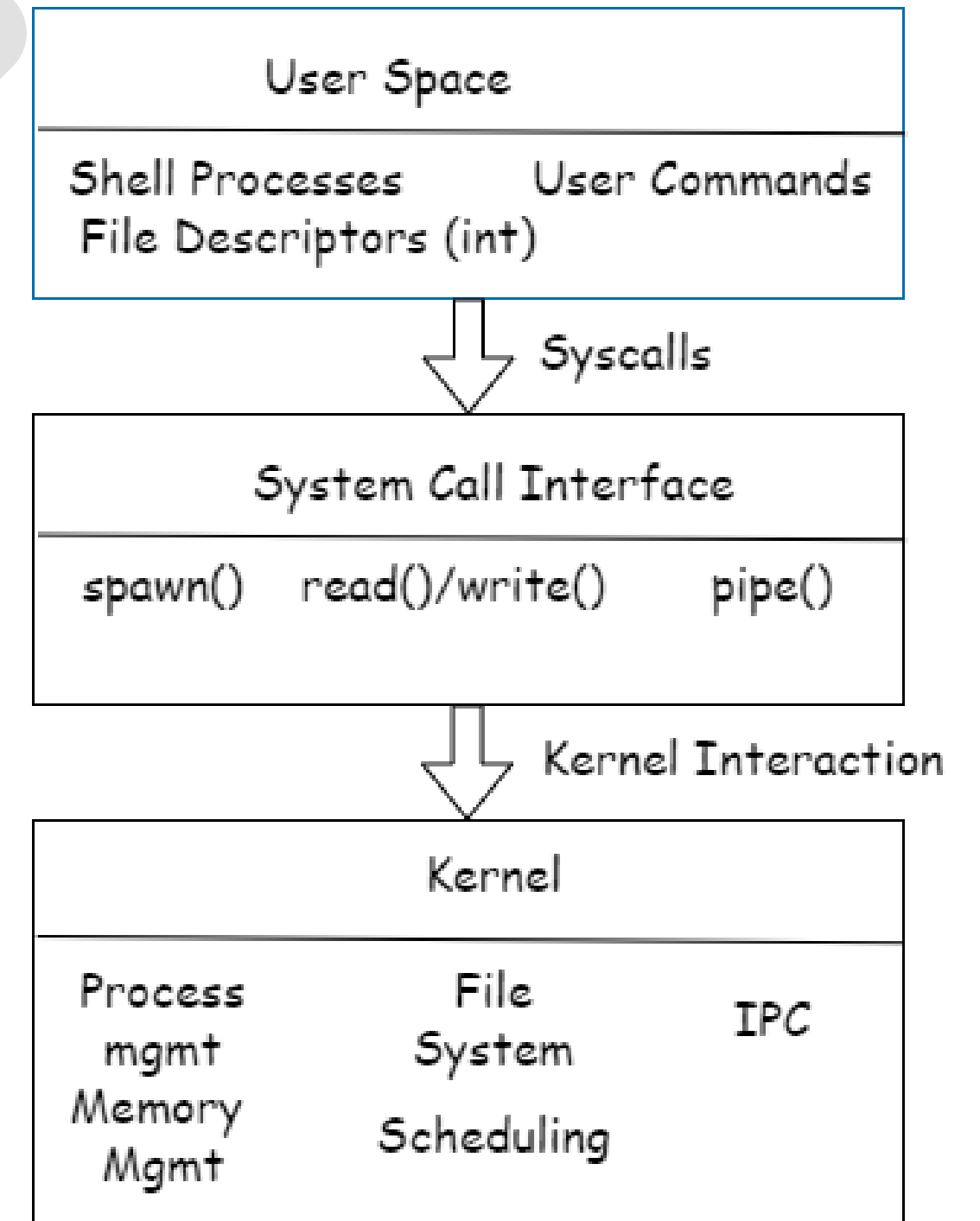
- Modify `spawn()` or create a new syscall (`exec()`) to load and execute ELF binaries from disk.
- Update the QEMU disk image to store and retrieve these programs dynamically.

LAYERS & ARCHITECTURE

User Space (Top Layer) –
Where the shell and
commands run

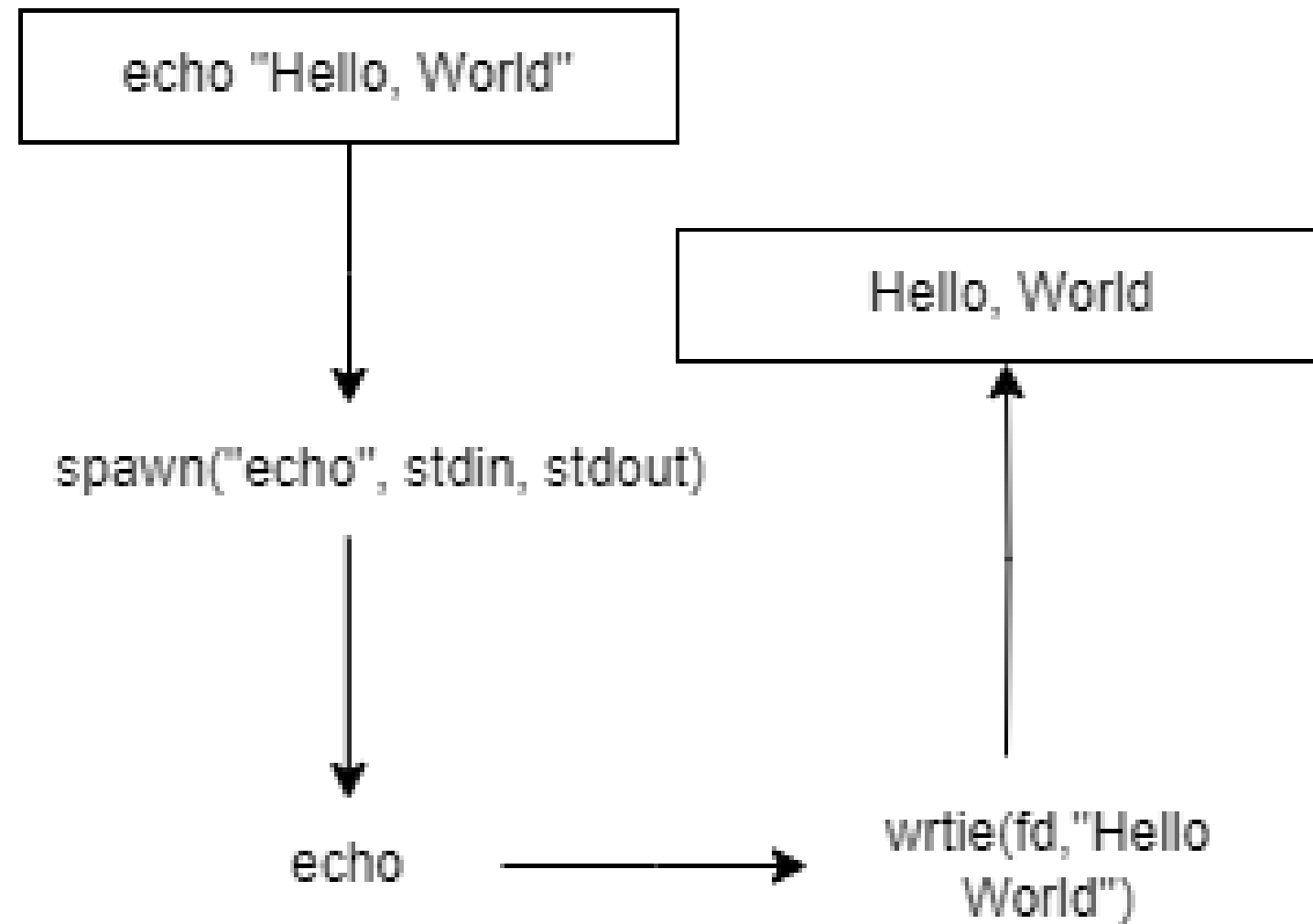
Syscalls connect User
Space to Kernel ↓

Kernel (Bottom Layer) –
Handles process
execution & resource
management



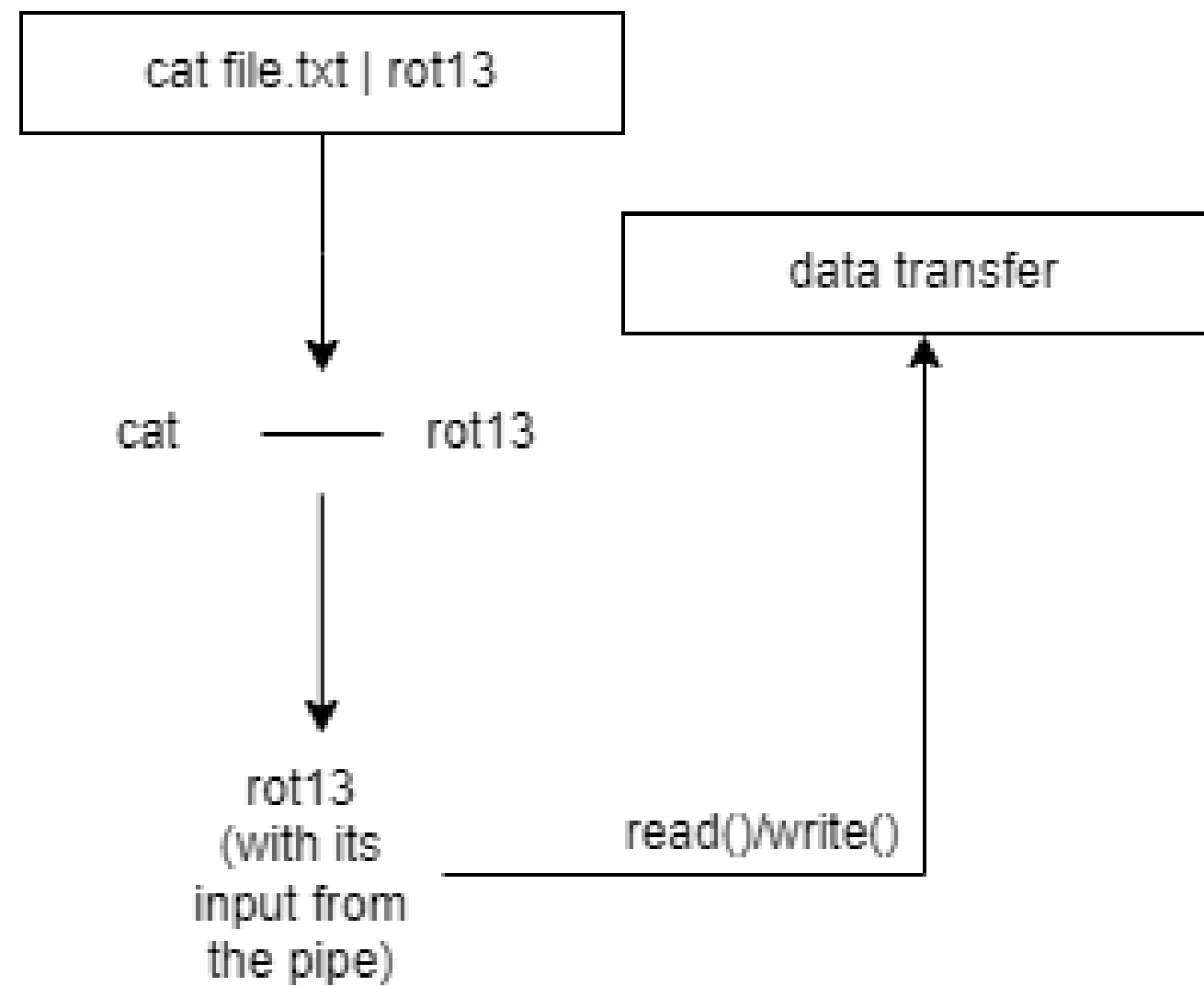
EXAMPLES

- RUNNING A SIMPLE COMMAND :



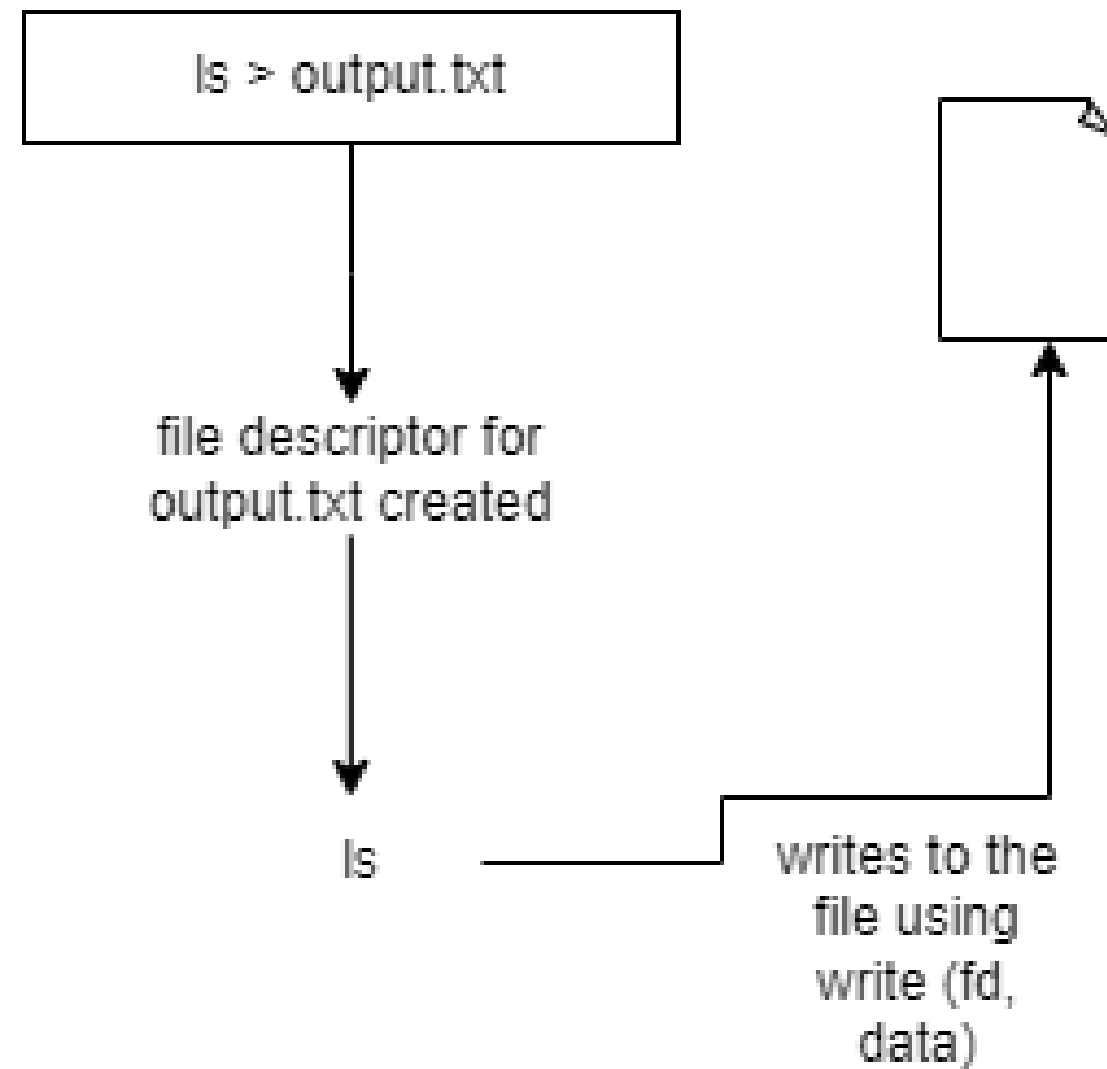
EXAMPLES

- USING PIPES :



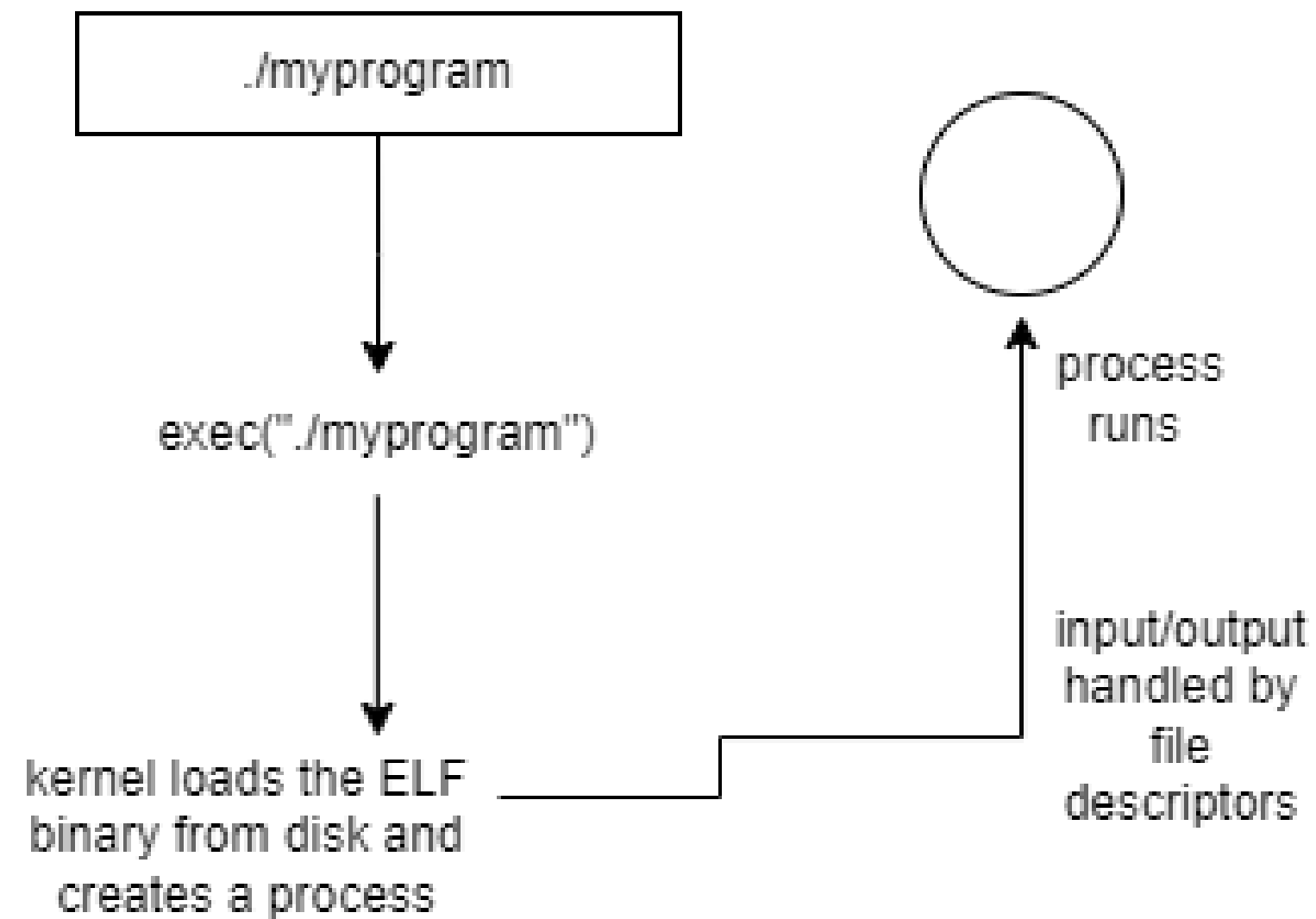
EXAMPLES

- FILE REDIRECTION (>,<)



EXAMPLES

- RUNNING A BIN FROM DISK :



CONCLUSION

AND FUTURE WORK

- EXPAND SYSTEM CALL APIS
- OPTIMIZE SHELL PERFORMANCE
- IMPROVE UX
- SECURITY ENHANCEMENTS



THANK YOU